

Spectral characteristics of brain activity during focused and mind-wandering states across frequency bands

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Abstract—Mind wandering is a common cognitive phenomenon characterized by a shift of attention away from the ongoing task toward internally generated thoughts. Its objective assessment using electroencephalography (EEG) remains challenging due to the complex and dynamic nature of brain activity. In this study, we propose an EEG-based framework for characterizing and classifying mind-wandering states using multiscale EEG features. Various Complexity feature extraction techniques are employed to obtain distinct oscillatory components from multichannel EEG signals, followed by extraction of spectral, statistical, entropy-based, and nonlinear features. The extracted features are subsequently evaluated using machine learning techniques to distinguish between focused and mind-wandering states. The proposed framework provides a multiscale representation of EEG dynamics and identifies informative features associated with changes in cognitive state. The findings demonstrate the potential of objective characterization of mind-wandering-related neural activity.

Keywords—Mind wandering, EEG, Spectral Features, Nonlinear Features, Cognitive State Classification.

I. INTRODUCTION

Mind wandering is a prevalent cognitive state in which attention shifts away from an ongoing task toward internally generated thoughts [1]. Previous research has established that mind wandering occurs regularly during everyday activities and is frequently accompanied by attentional lapses and diminished task performance [2]. From a neural perspective, internally directed cognition has been associated with the default mode network (DMN), providing a neurobiological foundation for spontaneous thought generation [3]. More recent theoretical frameworks describe mind wandering as a dynamic interplay between spontaneous thought generation and executive control processes [4]. While excessive mind wandering can negatively affect performance on ongoing tasks, it may also support cognitive functions such as creativity, prospective thinking, and autobiographical reflection [5]. Electroencephalography (EEG) has become an effective approach for examining the neural mechanisms associated with mind wandering due to its high temporal resolution and sensitivity to rapid fluctuations in brain activity [6]. Meditation, meanwhile, offers a valuable experimental context for examining attentional fluctuations because it directly engages sustained attention and meta-awareness processes [7], [8]. Previous neuroimaging and electrophysiological research has demonstrated

that meditation can produce measurable alterations in brain structure, functional activity, and neural oscillations across different frequency bands [9]–[11]. Consequently, meditation-based paradigms provide a promising framework for examining the neural dynamics underlying transitions between focused attention and mind wandering. Recent methodological developments have facilitated the direct examination of mind wandering during meditation through experience-sampling approaches. Notably, [12] utilized a probe-caught EEG paradigm to compare breath-focused attention with mind wandering in novice meditators. Their results demonstrated differential changes in alpha and theta activity, including variations in oscillatory amplitude, peak frequency, and cross-frequency coupling, emphasizing the role of these neural rhythms in internally directed cognitive processing. Despite these developments, existing EEG investigations of mind wandering have predominantly relied on conventional spectral analysis and predefined frequency bands. Nevertheless, EEG signals exhibit complex nonlinear, nonstationary, and multiscale characteristics [13], highlighting the need for adaptive signal-processing techniques that can reveal underlying oscillatory dynamics. The main contributions of this study are summarized as follows:

- **Physiological biomarkers of mind wandering:** We identify consistent spectral signatures of mind wandering across two EEG datasets. Delta power exhibits a highly consistent positive shift during mind wandering relative to focused attention ($\sim +0.095$ and $\sim +0.099 \log_{10}(\text{V}^2/\text{Hz})$ in Datasets 1 and 2, respectively), together with divergent temporal trajectories, decreasing during sustained focus and increasing during mind wandering. Theta power is elevated during mind wandering, with a significant effect in Dataset 2 ($+0.0488$, $p < 0.001$), while Gamma power shows consistent suppression (~ -0.032 and ~ -0.031). In contrast, Beta power does not exhibit a statistically significant state-dependent difference in either dataset.
- **Feature engineering and importance analysis:** We systematically quantify the contribution of different EEG feature families and demonstrate the dominance of spectral band-power features, accounting for 50.4% and 41.5% of total feature importance in Datasets 1 and 2,

respectively. Hjorth parameters demonstrate high information density, while phase-based functional connectivity contributes less than 3.2% of total importance despite constituting the largest feature group, indicating substantial redundancy of long-range phase-synchronization features.

- **Machine learning and preprocessing insights:** We compare conventional and nonlinear classification approaches and demonstrate the superiority of advanced classifiers, including Deep MLP and ExtraTrees, over standard linear baselines such as SVM and Logistic Regression. We further examine the effect of ICA-based artifact removal and show that its impact depends on the EEG montage and classifier, with negligible changes for the lower-density Dataset 1 and a decrease in SVM performance for the higher-density Dataset 2.
- **Spatial dynamics of attention:** We investigate the spatial distribution of predictive EEG information and identify Central–Temporal and Posterior scalp regions as the most informative spatial configurations, whereas frontal-only configurations provide comparatively weaker standalone discrimination. These findings provide evidence that the neural signatures associated with mind wandering are distributed across specific spatial regions rather than being exclusively localized to frontal electrodes.

II. RELATED WORK

Mind wandering has become an important research area in cognitive neuroscience due to its close relationship with attention, memory, and executive control processes. The increasing use of electroencephalography has allowed researchers to examine the temporal characteristics of spontaneous thought with high temporal resolution. A comprehensive review by [6] examined forty-two EEG studies and identified attenuated event-related responses and elevated low-frequency oscillatory activity, particularly within the theta and alpha bands, as the most consistent electrophysiological markers of mind wandering. The review also reported considerable variability in spectral findings across experimental paradigms, underscoring the necessity for more reliable and comprehensive analytical approaches. Meditation-based paradigms offer a suitable experimental framework for examining mind wandering, as they naturally involve transitions between externally focused attention and internally generated thoughts. Mindfulness training has been consistently associated with reductions in task-unrelated thoughts and alterations in default mode network activity [14]. A systematic review by [15] synthesized findings from behavioral and neurophysiological investigations and reported that experienced meditators demonstrate fewer instances of mind wandering and improved attentional regulation compared with novice practitioners. Furthermore, neuroimaging research has indicated that different meditation practices engage distinct neural networks associated with attention, self-referential processing, and cognitive control [7], [10], [16]. In this context, [12] developed an experience-sampling EEG paradigm to directly compare breath-focused attention with mind wander-

ing during meditation. Their results showed increased theta amplitude, reduced alpha amplitude, alterations in oscillatory peak frequencies, and stronger alpha-theta coupling during mind-wandering episodes. In a subsequent investigation, [17] compared novice meditators with experienced practitioners and identified significant differences in EEG spectral dynamics during meditation and spontaneous thought, suggesting that meditation expertise affects the neural processes associated with attentional fluctuations. Extending conventional spectral investigations, [18] incorporated nonlinear EEG measures and observed reduced neural complexity during mind wandering compared with breath-focused attention. These findings indicate that conventional spectral characteristics may not fully represent the complex neural dynamics involved in internally directed cognition. Consistent findings have emerged from meditation research utilizing entropy, complexity, and connectivity measures, demonstrating that meditation can alter large-scale neural interactions across multiple frequency ranges [19]. Recent reviews have further highlighted the increasing relevance of EEG-derived biomarkers for investigating the neural mechanisms underlying mindfulness and meditation. Alongside these neuroscientific investigations, substantial research has focused on the automated identification of mind wandering through machine-learning techniques. Initial studies largely utilized handcrafted spectral features derived from canonical EEG frequency bands together with traditional classifiers, including support vector machines, random forests, and multilayer perceptrons [20], [21]. Subsequently, deep-learning methods, particularly convolutional neural networks, have been investigated for performing end-to-end mind-wandering classification directly from EEG recordings [22]. Broader investigations of deep learning for EEG analysis have demonstrated increasing adoption of convolutional and recurrent architectures for decoding cognitive states, while also identifying challenges related to reproducibility and the interpretability of learned features. Despite these developments, most existing investigations continue to characterize EEG activity either directly in the time domain or using predefined frequency bands. Given the nonlinear and nonstationary nature of EEG signals, adaptive signal-processing methods are increasingly relevant for identifying latent oscillatory components.

III. PROPOSED METHODOLOGY

A. Dataset

We employed the publicly available dataset provided by the authors of [23]. In this study, breath counting was used to represent the non-mind or breath focus (BF) state. The dataset comprises recordings from two healthy participants (one male and one female), each of whom completed eleven experimental sessions designed to capture episodes of mind wandering. During the experiment, participants performed a reverse breath-counting task from 10 to 1 while fixating on a cross displayed on a computer screen. Whenever they became aware that their attention had shifted away from the task, they pressed a button to indicate the onset of a mind-wandering (MW) episode, completed a questionnaire

describing the experience, and subsequently pressed another button to indicate refocusing before resuming the counting task. Each session lasted approximately 20 minutes. EEG signals were acquired using a 64-channel BIOSEMI recording system at a sampling frequency of 1024 Hz. The recorded signals were subsequently band-pass filtered between 0.5 and 40 Hz. Following preprocessing and artifact removal, a total of 977 EEG epochs were extracted, including 472 MW epochs and 505 breath-focused (BF) epochs. Similarly another dataset from [12], acquired using the Nexus-32 recording system using a 22-electrode cap, comprising 19 scalp electrodes, two reference electrodes, and one ground electrode, positioned according to the international 10–20 system. EEG signals were amplified using a unipolar amplifier and sampled at 512 Hz. To monitor ocular activity, vertical electrooculogram (VEOG) and horizontal electrooculogram (HEOG) signals were recorded using pre-gelled foam electrodes placed above and below the left eye (VEOG) and at the outer canthi of both eyes (HEOG), with a sampling rate of 2048 Hz.

B. Preprocessing

To ensure a consistent and comparable analysis, both EEG datasets were subjected to dataset specific preprocessing before feature extraction and classification. For Dataset 1, the raw EEG recordings contained 21 channels, of which 19 were retained as EEG channels, while the remaining two channels were treated as auxiliary channels. The recordings were converted from microvolts to volts, assigned a standard 10–20 montage, and band-pass filtered between 1 and 40 Hz using a FIR filter. Channels exhibiting extremely low variance were identified as bad channels and subsequently interpolated using spherical interpolation, followed by average referencing. The behavioral event markers were used to identify Focus and mind-wandering states, and EEG segments preceding each event were extracted. A 4.9-s segment (–5 to –0.1 s) was extracted for both conditions and zero-duration differences were compensated by edge padding to obtain uniform 5-s epochs. For the artifact-cleaned version of Dataset 1, the same initial filtering, channel handling, interpolation, referencing, and epoching procedure was retained. The simultaneously recorded two-channel EOG signals were independently band-pass filtered between 1 and 40 Hz, segmented using the corresponding event markers, and resampled to the EEG sampling rate. ICA was then performed on the epoched EEG data using the FastICA algorithm. The resulting independent components were correlated with the EOG channels using Pearson correlation, and components showing an absolute correlation greater than 0.3 with either EOG channel were identified as ocular-artifact components and removed before reconstructing the cleaned EEG epochs. For Dataset 2, the recordings consisted of 64 EEG channels together with eight auxiliary EXG channels. The EEG and EXG channels were first identified from the BIDS channel metadata, after which the recordings were resampled to 256 Hz. The event markers were correspondingly adjusted to maintain temporal alignment following resampling. The recordings were then band-pass filtered between 0.5 and

45 Hz. Two matched signal conditions were subsequently generated from the same filtered recordings: Raw, containing the 64 EEG channels without ICA-based artifact removal, and ICA-Cleaned, in which ICA was applied to the EEG channels and components showing an absolute correlation of ≥ 0.30 with the auxiliary EXG channels were removed. For Dataset 2, event markers were used to construct the analysis windows for the two cognitive states. Mind-wandering epochs were extracted from –5 to –0.1 s relative to the mind-wandering report, whereas Focus epochs were extracted from +1 to +6 s relative to the corresponding task event. The resulting epochs were standardized to a common 5-s duration, with the shorter mind-wandering segments edge-padded where required. Thus, for both datasets, identical downstream feature extraction and classification procedures were applied separately to the raw and ICA-cleaned conditions, enabling a direct assessment of the influence of artifact removal on mind-wandering classification performance.

C. Feature Extraction

Following preprocessing and epoching, a comprehensive set of 18 features was extracted independently from each EEG channel to characterize the temporal, statistical, spectral, and complexity-related properties of the neural signals. The same feature extraction procedure was applied to both the raw and ICA-cleaned EEG conditions to ensure a direct comparison of the information available before and after artifact removal. The extracted features comprised six statistical/time-domain features (Mean, Variance, Standard Deviation, Root Mean Square (RMS), Kurtosis, and Skewness), three Hjorth parameters (Activity, Mobility, and Complexity), Peak-to-Peak amplitude, five spectral features (Total Power, Theta Power, Alpha Power, Spectral Entropy, and Dominant Frequency), and three additional signal-complexity/energy measures (Energy, Shannon Entropy, and Differential Entropy). Thus, the feature set captured both conventional amplitude characteristics and frequency- and complexity-based variations associated with the two cognitive states. For spectral characterization, the Welch power spectral density (PSD) method was employed. Total power was calculated over the analyzed frequency range, while band-specific power was obtained for the theta (4–8 Hz) and alpha (8–12 Hz) bands. Spectral entropy was calculated from the normalized PSD distribution to quantify the spectral complexity of each epoch, and the dominant frequency was identified as the frequency corresponding to the maximum PSD value. The Hjorth parameters were calculated from the signal and its first- and second-order differences, providing measures of signal activity, mobility, and complexity. Shannon entropy was estimated from the amplitude histogram, while differential entropy was calculated from the signal variance. The features were extracted channel-wise for every 5-s epoch and subsequently concatenated to form the feature vector for classification. For each epoch, the corresponding cognitive-state label was retained along with subject, session, epoch, and artifact-condition information. Consequently, Dataset 1 generated $18 \text{ features} \times 19 \text{ EEG channels} = 342 \text{ features}$

per epoch, whereas Dataset 2 generated 18 features $\times$ 64 EEG channels = 1,152 features per epoch. This channel-wise representation preserved the spatial information of the EEG while providing a unified feature space for the subsequent machine-learning analyses.

D. Feature-wise Importance and Band-wise Analysis

To quantify the contribution of individual EEG characteristics to the discrimination between Focus and mind-wandering states, a feature-wise analysis based on an Extra Trees ensemble was performed separately for the raw and ICA-cleaned EEG conditions. Let the preprocessed EEG signal for an epoch be represented as

$$\mathbf{X} \in \mathbb{R}^{C \times T}, \quad (1)$$

where C denotes the number of EEG channels and T denotes the number of temporal samples. Each epoch was divided into overlapping temporal windows of length L with a step size S . Thus, the w -th window is given by

$$\mathbf{X}^{(w)} = \mathbf{X}[:, (w-1)S : (w-1)S + L]. \quad (2)$$

For Dataset 1, $L = 2$ s and $S = 0.5$ s, resulting in six overlapping windows per 5-s epoch. For Dataset 2, $L = 1.5$ s and $S = 0.5$ s were used.

For each window and channel, the signal was decomposed into five frequency bands:

$$\mathcal{B} = \{\delta, \theta, \alpha, \beta, \gamma\}, \quad (3)$$

where

$$\delta = [1, 4) \text{ Hz}, \quad \theta = [4, 8) \text{ Hz}, \quad \alpha = [8, 12) \text{ Hz}, \quad (4)$$

$$\beta = [13, 30) \text{ Hz}, \quad \gamma = [30, 45] \text{ Hz}. \quad (5)$$

For a channel signal $x_c(t)$, the band-limited signal was obtained using a fourth-order Butterworth band-pass filter,

$$x_{c,b}(t) = \mathcal{F}_b\{x_c(t)\}, \quad (6)$$

where $\mathcal{F}_b$ denotes the band-pass filtering operation corresponding to frequency band b .

The power spectral density of each window was estimated and the absolute band power was calculated as

$$P_{c,b} = \int_{f_{b,l}}^{f_{b,h}} PSD_c(f) df. \quad (7)$$

The corresponding relative band power was defined as

$$P_{c,b}^{rel} = \frac{P_{c,b}}{\sum_{f=1}^{45} PSD_c(f) + \epsilon}, \quad (8)$$

where ϵ is a small positive constant introduced to avoid numerical instability.

For the theta and alpha bands, the analytic signal was obtained using the Hilbert transform:

$$z_{c,b}(t) = x_{c,b}(t) + j\mathcal{H}\{x_{c,b}(t)\}, \quad (9)$$

and its amplitude envelope was calculated as

$$A_{c,b}(t) = |z_{c,b}(t)|. \quad (10)$$

The mean envelope, variance, coefficient of variation, and burst count were subsequently calculated. The coefficient of variation was defined as

$$CV_{c,b} = \frac{\sigma(A_{c,b})}{\mu(A_{c,b}) + \epsilon}. \quad (11)$$

A burst was defined using a threshold of

$$\tau_{c,b} = \mu(A_{c,b}) + 2\sigma(A_{c,b}), \quad (12)$$

and the burst count was obtained by counting upward threshold crossings.

In addition to band-specific features, alpha-envelope synchronization between channels was calculated using the Pearson correlation coefficient. For channels i and j ,

$$R_{ij}^\alpha = \text{corr}(A_{i,\alpha}(t), A_{j,\alpha}(t)). \quad (13)$$

The synchronization associated with channel i was then obtained by averaging its correlation with all other channels:

$$S_i^\alpha = \frac{1}{C-1} \sum_{\substack{j=1 \\ j \neq i}}^C R_{ij}^\alpha. \quad (14)$$

Consequently, each temporal window was represented by a high-dimensional feature vector

$$\mathbf{f}^{(w)} = [f_1, f_2, \dots, f_D]^T, \quad (15)$$

where D denotes the total number of channel-wise and band-wise features. The features from all EEG channels were concatenated while retaining their channel identity, producing the final input matrix

$$\mathbf{F} \in \mathbb{R}^{N \times D}, \quad (16)$$

where N represents the number of EEG windows and D represents the total number of extracted features.

1) *Extra Trees Feature Importance*: An Extra Trees classifier was employed to estimate the relative contribution of the extracted features. Let the ensemble contain M decision trees,

$$\mathcal{E} = \{T_1, T_2, \dots, T_M\}. \quad (17)$$

For each feature f_j , the importance obtained from the ensemble is represented by

$$I_j = \frac{1}{M} \sum_{m=1}^M I_j^{(m)}, \quad (18)$$

where $I_j^{(m)}$ denotes the feature importance of feature j in tree m . The importance values are normalized by the Extra Trees implementation such that

$$\sum_{j=1}^D I_j = 1. \quad (19)$$

For classification, the model was evaluated using stratified group-wise cross-validation. The grouping variable was the

subject-session identifier, ensuring that samples originating from the same recording session were not simultaneously used for training and testing. The out-of-fold predictions were used to calculate Accuracy, Precision, Recall, F1-score, and ROC-AUC.

For the descriptive feature-importance analysis, an Extra Trees model was subsequently fitted to the complete feature set. The resulting importance values were ranked to identify the most influential individual EEG features. This full-data importance model was used only to characterize feature contribution and was not used as an estimate of generalization performance.

2) *Band-wise Feature Importance*: To determine the relative contribution of different frequency ranges, individual feature importance values were aggregated according to their associated frequency band. Let $\mathcal{J}_b$ denote the set of features belonging to frequency band b . The total importance of band b was calculated as

$$I_b^{total} = \sum_{j \in \mathcal{J}_b} I_j. \quad (20)$$

The resulting band-wise importance therefore satisfies

$$\sum_{b \in \mathcal{B}} I_b^{total} + I_{Other} = 1, \quad (21)$$

where I_{Other} represents features that cannot be uniquely assigned to one of the predefined frequency bands.

This analysis provides two complementary levels of interpretation: the individual feature importance I_j identifies the specific EEG characteristics that contribute most strongly to Focus versus mind-wandering discrimination, whereas I_b^{total} quantifies the aggregate contribution of each frequency band. The analysis was performed independently for the raw and ICA-cleaned conditions, enabling examination of whether artifact removal altered the relative importance of spectral and temporal EEG characteristics.

E. Band-wise Temporal Analysis

To investigate the spectral characteristics associated with Focus and mind-wandering states, a band-wise temporal analysis was performed on the ICA-cleaned EEG recordings. Let the EEG signal corresponding to an epoch be represented as

$$\mathbf{X} \in \mathbb{R}^{C \times T}, \quad (22)$$

where C denotes the number of EEG channels and T denotes the number of temporal samples. Each 5-s epoch was divided into overlapping temporal windows of length $L = 1$ s with a step size of $S = 0.5$ s. For each window, the power spectral density (PSD) was estimated using Welch's method.

The analysis considered five conventional EEG frequency bands:

$$\mathcal{B} = \{\delta, \theta, \alpha, \beta, \gamma\}, \quad (23)$$

with frequency ranges

$$\delta = [1, 4) \text{ Hz}, \quad \theta = [4, 8) \text{ Hz}, \quad \alpha = [8, 12) \text{ Hz}, \quad (24)$$

$$\beta = [13, 30) \text{ Hz}, \quad \gamma = [30, 45] \text{ Hz}. \quad (25)$$

For a given window w , channel c , and frequency band b , the band power was calculated from the estimated PSD as

$$P_{c,b}^{(w)} = \int_{f_{b,l}}^{f_{b,h}} PSD_c^{(w)}(f) df. \quad (26)$$

The logarithmic representation of band power was then obtained as

$$L_{c,b}^{(w)} = \log_{10} \left(P_{c,b}^{(w)} + \epsilon \right), \quad (27)$$

where ϵ is a small positive constant used to avoid numerical instability.

For each temporal window, the channel-wise band powers were averaged across the available EEG channels to obtain a representative band-power measure:

$$L_b^{(w)} = \frac{1}{C} \sum_{c=1}^C L_{c,b}^{(w)}. \quad (28)$$

This resulted in a temporal trajectory of logarithmic band power for each EEG frequency band.

To obtain an epoch-level representation, the mean band power across all valid windows within an epoch was calculated as

$$\bar{L}_{b,e} = \frac{1}{W_e} \sum_{w=1}^{W_e} L_{b,e}^{(w)}, \quad (29)$$

where W_e denotes the number of valid windows in epoch e . In addition to the mean power, the temporal evolution of each frequency band was quantified by fitting a first-order linear model to the window-level power:

$$L_{b,e}^{(w)} = \beta_{0,b,e} + \beta_{1,b,e} t_w + \varepsilon_{b,e}^{(w)}, \quad (30)$$

where t_w represents the center time of window w and $\beta_{1,b,e}$ represents the within-epoch rate of change in band power. Thus, a positive $\beta_{1,b,e}$ indicates an increase in band power over the epoch, whereas a negative value indicates a decrease.

The difference in mean band power between mind-wandering and Focus states was evaluated using an ordinary least-squares model:

$$\bar{L}_{b,e} = \beta_{0,b} + \beta_{1,b} I_{MW,e} + \varepsilon_e, \quad (31)$$

where $I_{MW} = 1$ for mind-wandering epochs and $I_{MW} = 0$ for Focus epochs. Consequently, $\beta_{1,b}$ represents the estimated difference in mean logarithmic band power between mind-wandering and Focus states:

$$\Delta_b = \bar{L}_{b,MW} - \bar{L}_{b,Focus}. \quad (32)$$

To characterize state-specific temporal changes, a second regression model was fitted to the epoch-level slopes:

$$\beta_{1,b,e} = \gamma_{0,b} + \gamma_{1,b} I_{MW,e} + \eta_e. \quad (33)$$

Here, $\gamma_{0,b}$ represents the temporal change in band power during Focus, whereas $\gamma_{0,b} + \gamma_{1,b}$ represents the corresponding temporal change during mind wandering. Statistical inference was performed using subject-clustered standard errors to account for the repeated observations obtained from the same

individual. The resulting estimates were reported together with their standard errors, 95% confidence intervals, and two-sided p -values. Statistical significance was assessed at $\alpha = 0.05$.

F. Raw versus ICA-Cleaned Classification

To quantify the influence of artifact removal on the discrimination of Focus and mind-wandering states, classification was independently performed on two signal conditions: Raw EEG and ICA-Cleaned EEG. The two conditions were generated from the same recordings and analyzed using identical feature-extraction and classification procedures. This paired design ensured that differences in classification performance could be attributed to the effect of artifact removal rather than differences in the underlying recording samples.

Let the feature representation of an EEG epoch be denoted by

$$\mathbf{x}_i \in \mathbb{R}^D, \quad (34)$$

where D is the total number of extracted channel-wise features and $y_i \in \{0, 1\}$ denotes the corresponding cognitive state, with 0 representing Focus and 1 representing mind wandering. Separate feature matrices were constructed for the Raw and ICA-Cleaned conditions,

$$\mathbf{X}^{(R)}, \mathbf{X}^{(C)} \in \mathbb{R}^{N \times D}, \quad (35)$$

where R and C denote Raw and ICA-Cleaned EEG, respectively.

Prior to classification, missing feature values were replaced using median imputation. The resulting features were standardized within the classification pipeline according to

$$z_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \quad (36)$$

where μ_j and σ_j represent the mean and standard deviation of feature j estimated from the corresponding training data. The imputation and standardization operations were performed as part of the model pipeline to avoid using information from the test folds during preprocessing.

IV. MODEL TRAINING

Two conventional supervised classifiers were evaluated: a nonlinear Support Vector Machine (SVM) and Logistic Regression (LR). For the SVM, the radial basis function (RBF) kernel was employed:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2), \quad (37)$$

where γ controls the width of the RBF kernel. The SVM was configured with $C = 1$ and $\gamma = \text{scale}$, together with balanced class weighting.

For Logistic Regression, the probability of the mind-wandering class was modeled as

$$P(y_i = 1 | \mathbf{x}_i) = \frac{1}{1 + \exp(-(\mathbf{w}^T \mathbf{x}_i + b))}, \quad (38)$$

where $\mathbf{w}$ is the vector of model coefficients and b is the intercept. Balanced class weighting was also employed for Logistic Regression.

To reduce the possibility of information leakage caused by repeated observations from the same individual, grouped cross-validation was employed. Let g_i denote the group identifier associated with sample i . The data were partitioned into K folds such that

$$g_i \in \mathcal{G}_k \Rightarrow i \notin \mathcal{T}_l, \quad l \neq k, \quad (39)$$

thereby ensuring that samples belonging to the same group were not independently distributed across training and testing folds. Stratified Group K-Fold cross-validation with shuffling and a fixed random seed was used whenever possible.

For each fold k , the classifier was trained using the training subset,

$$f^{(k)} : \mathbf{X}_{train}^{(k)} \rightarrow y_{train}^{(k)}, \quad (40)$$

and predictions were generated for the held-out subset,

$$\hat{y}_{test}^{(k)} = f^{(k)}(\mathbf{X}_{test}^{(k)}). \quad (41)$$

The predictions from all held-out folds were subsequently combined to obtain a complete set of out-of-fold predictions. Thus, every valid sample received a prediction from a model that had not been trained using that sample.

For ROC-AUC estimation, the continuous decision score generated by the classifier was retained rather than using only the binary class prediction. For the SVM, the signed decision function was used, whereas the corresponding class-probability/decision score was used where available. The ROC-AUC was then calculated from the out-of-fold decision scores and the corresponding ground-truth labels.

The complete analysis was performed independently for the Raw and ICA-Cleaned conditions. The difference in performance between the two conditions was therefore quantified for each classifier and metric as

$$\Delta M = M_{ICA} - M_{Raw}, \quad (42)$$

where M represents Accuracy, Precision, Recall, F1-score, or ROC-AUC. A negative value of ΔM indicates a reduction in classification performance following ICA cleaning, whereas a positive value indicates an improvement.

1) *Deep Multilayer Perceptron*: To investigate whether the extracted high-dimensional EEG representation could be effectively exploited by a nonlinear deep-learning model, a Deep Multilayer Perceptron (DeepMLP) was employed for binary classification of Focus and mind-wandering states. The DeepMLP was evaluated independently on the Raw and ICA-Cleaned EEG conditions using the same feature representation and epoch-level evaluation framework.

Let $\mathbf{x} \in \mathbb{R}^D$ denote the feature vector corresponding to an EEG window, where D is the dimensionality of the extracted feature representation. The DeepMLP consists of a sequence of fully connected layers with batch normalization, GELU nonlinear activation, and dropout regularization. For a hidden layer l , the transformation can be expressed as

$$\mathbf{h}_l = \mathcal{D}_{p_l} [\text{GELU}(\text{BN}(\mathbf{W}_l \mathbf{h}_{l-1} + \mathbf{b}_l))], \quad (43)$$

where $\mathbf{W}_l$ and $\mathbf{b}_l$ denote the weight matrix and bias vector of layer l , $\text{BN}(\cdot)$ represents batch normalization, $\text{GELU}(\cdot)$ is the Gaussian Error Linear Unit activation function, and $\mathcal{D}_{p_l}(\cdot)$ denotes dropout with probability p_l .

For Dataset 2, the network architecture was

$$D \rightarrow 1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 1, \quad (44)$$

with dropout probabilities of 0.50, 0.40, 0.30, and 0.20 after the respective hidden layers. The final layer produces a scalar logit,

$$z = \mathbf{W}_o \mathbf{h}_L + b_o, \quad (45)$$

which is converted into the probability of the mind-wandering class using the sigmoid function:

$$\hat{p} = \sigma(z) = \frac{1}{1 + \exp(-z)}. \quad (46)$$

The network was optimized using the weighted binary cross-entropy loss with logits:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N [w_+ y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)], \quad (47)$$

where $y_i \in \{0, 1\}$ denotes the ground-truth class and w_+ represents the positive-class weight calculated from the class distribution of the training fold. This weighting was employed to compensate for class imbalance. The network parameters were optimized using the AdamW optimizer with a learning rate of 10^{-3} and weight decay of 10^{-4} . A ReduceLROnPlateau learning-rate scheduler was employed to reduce the learning rate when the evaluation loss ceased to improve. Training was terminated using early stopping with a patience of 25 epochs, while the model state corresponding to the lowest evaluation loss was retained. The maximum number of training epochs was set to 150 with a batch size of 64. Classification performance was quantified using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Accuracy was calculated as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (48)$$

while Precision and Recall were defined as

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (49)$$

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (50)$$

The F1-score was calculated as the harmonic mean of Precision and Recall:

$$F1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (51)$$

V. RESULTS AND DISCUSSION

The band-wise analysis revealed several consistent spectral patterns across the two EEG datasets. Theta power showed a positive shift during mind-wandering relative to focused states in both datasets, with a stronger and statistically significant effect in Dataset 2 ($\hat{\beta} = 0.0488$, $p < 0.001$), while Dataset 1 exhibited a weaker positive trend ($\hat{\beta} = 0.0816$, $p = 0.1775$). Delta power also showed remarkably similar positive mean differences between mind-wandering and focus across Dataset 1 and Dataset 2 ($\hat{\beta} = 0.0948$ and 0.0992 , respectively). Furthermore, the temporal trajectories of Delta power exhibited opposite directions for the two cognitive states, decreasing during focus and increasing during mind-wandering. Gamma power showed a consistent suppression during mind-wandering in both datasets, with closely matched effect estimates of -0.0322 and -0.0305 , respectively. Overall, these convergent patterns suggest that low-frequency dynamics and Gamma suppression provide common spectral signatures of transitions between externally focused and internally oriented cognitive states. Across both datasets, the advanced classifiers consistently outperformed the conventional linear baselines, indicating that the distinction between focused and mind-wandering states is better captured through nonlinear feature interactions. Feature-family analysis further revealed a consistent dominance of band-power features, accounting for 50.4% and 41.5% of the total importance in Dataset 1 and Dataset 2, respectively. In contrast, phase-based functional connectivity contributed only 2.4% and 3.2% despite comprising 361 and 4096 features, respectively, suggesting substantial redundancy in long-range phase synchronization features. Hjorth parameters also exhibited high information density, supporting the relevance of localized signal complexity and temporal dynamics for state discrimination. Interestingly, Beta-band power exhibited a consistent negative but non-significant difference between mind-wandering and focused states in both datasets ($\hat{\beta} = -0.0494$, $p = 0.4445$ for Dataset 1; $\hat{\beta} = -0.0094$, $p = 0.5852$ for Dataset 2). This consistency suggests that Beta power alone is not a reliable standalone marker of state transitions across the two experimental settings. The ablation analysis revealed that the contribution of feature domains and spatial regions varied across datasets. In Dataset 1, isolating Theta features largely preserved baseline performance, while Central-Temporal electrodes provided the strongest standalone spatial representation. In Dataset 2, Alpha emerged as the dominant frequency band, with its omission substantially reducing SVM performance, whereas Central-Temporal and Posterior regions showed the strongest standalone spatial discriminability. These findings indicate that spectral and spatial information contribute differently depending on the EEG montage and experimental setting.

VI. CONCLUSION

This study investigated the neural signatures of mind wandering using EEG data from two complementary datasets with different recording densities and experimental settings. The results demonstrated several consistent physiological patterns

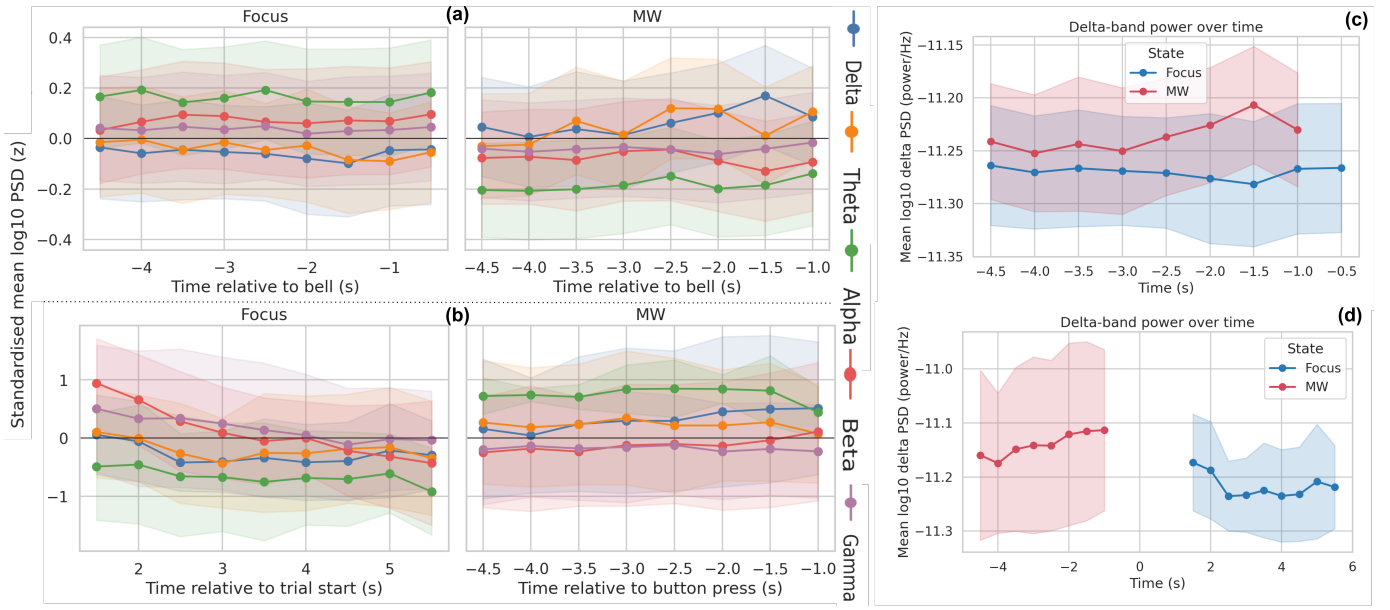


Fig. 1. bAND WISE tEMP aNALYSIS

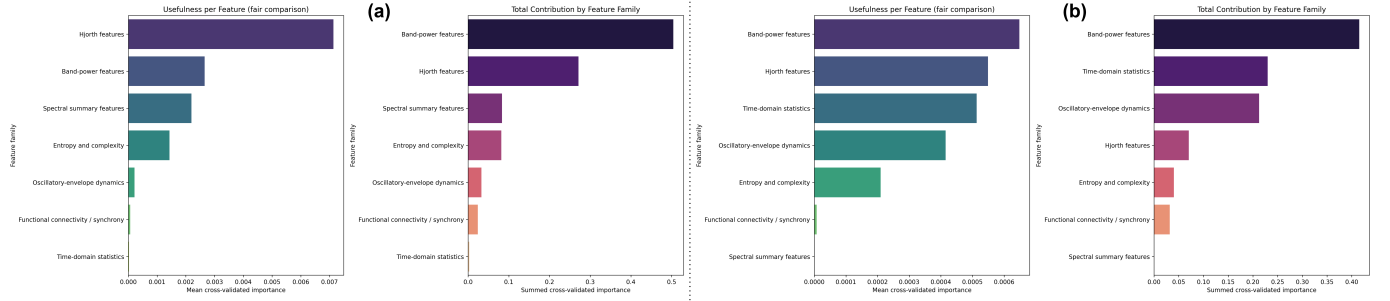


Fig. 2. to be decided

TABLE I
CLASSIFICATION PERFORMANCE OF SVM, LOGISTIC REGRESSION, AND DEEP MLP ON DATASET 1 AND DATASET 2. RESULTS ARE REPORTED FOR RAW AND ICA-CLEANED EEG.

Dataset	Condition	Model	Accuracy	Precision	Recall	F1	ROC-AUC
Dataset 1	Raw	SVM	60.40	54.46	54.71	54.59	63.33
		Logistic Regression	58.40	52.01	56.55	54.19	59.36
		Deep MLP	67.10	62.27	61.84	62.05	71.57
	ICA-cleaned	SVM	60.20	54.08	56.32	55.18	63.22
		Logistic Regression	58.00	51.53	57.93	54.55	60.31
		Deep MLP	67.40	63.46	59.08	61.19	71.44
Dataset 2	Raw	SVM	66.86	60.72	75.00	67.11	73.24
		Logistic Regression	67.34	63.32	65.47	64.38	72.15
		Deep MLP	85.39	84.90	82.20	83.53	93.71
	ICA-cleaned	SVM	62.27	55.90	77.33	64.89	66.36
		Logistic Regression	68.10	66.05	60.17	62.97	72.44
		Deep MLP	83.29	83.07	79.03	81.00	91.24

across datasets, including elevated Theta and Delta activity during mind wandering and a suppression of Gamma power, while Beta power did not emerge as a significant standalone marker. The temporal analysis further revealed divergent Delta trajectories, with power decreasing during sustained focus but increasing during mind wandering, suggesting progressive changes in vigilance and internally directed processing.

TABLE II
COMPACT ABILATION ANALYSIS USING SVM ON RAW EEG. THE BASELINE IS COMPARED WITH THE MOST INFORMATIVE AND LEAST INFORMATIVE VARIANTS WITHIN EACH ABILATION CATEGORY.

Category	Variant	Dataset 1			Dataset 2		
		Acc.	F1	AUC	Acc.	F1	AUC
Baseline	Full feature set	60.40	54.59	63.33	66.86	67.11	73.24
Feature family	Best omission	60.20	55.38	63.34	69.82	67.95	74.80
	Most damaging omission	59.30	53.59	62.52	60.17	53.72	68.23
Feature family	Best isolated	60.60	55.63	62.23	69.34	64.45	72.41
	Most damaging isolated	54.60	53.77	57.72	53.87	42.98	59.08
Scalp region	Best omission	61.30	55.67	64.16	71.44	68.95	77.35
	Most damaging omission	58.00	52.38	60.71	65.62	65.25	69.90
Frequency band	Best omission	59.72	54.58	62.54	65.94	59.06	68.96
	Most damaging omission	58.32	52.35	61.25	59.32	55.94	62.30
Frequency band	Best isolated	59.48	51.97	60.24	63.97	54.32	66.03
	Most damaging isolated	54.57	52.46	56.91	53.85	44.81	55.24

The machine learning and feature analyses further showed that nonlinear models, particularly the Deep MLP, provided stronger discrimination than conventional linear classifiers, highlighting the importance of complex interactions among EEG features. Band-power features constituted the dominant

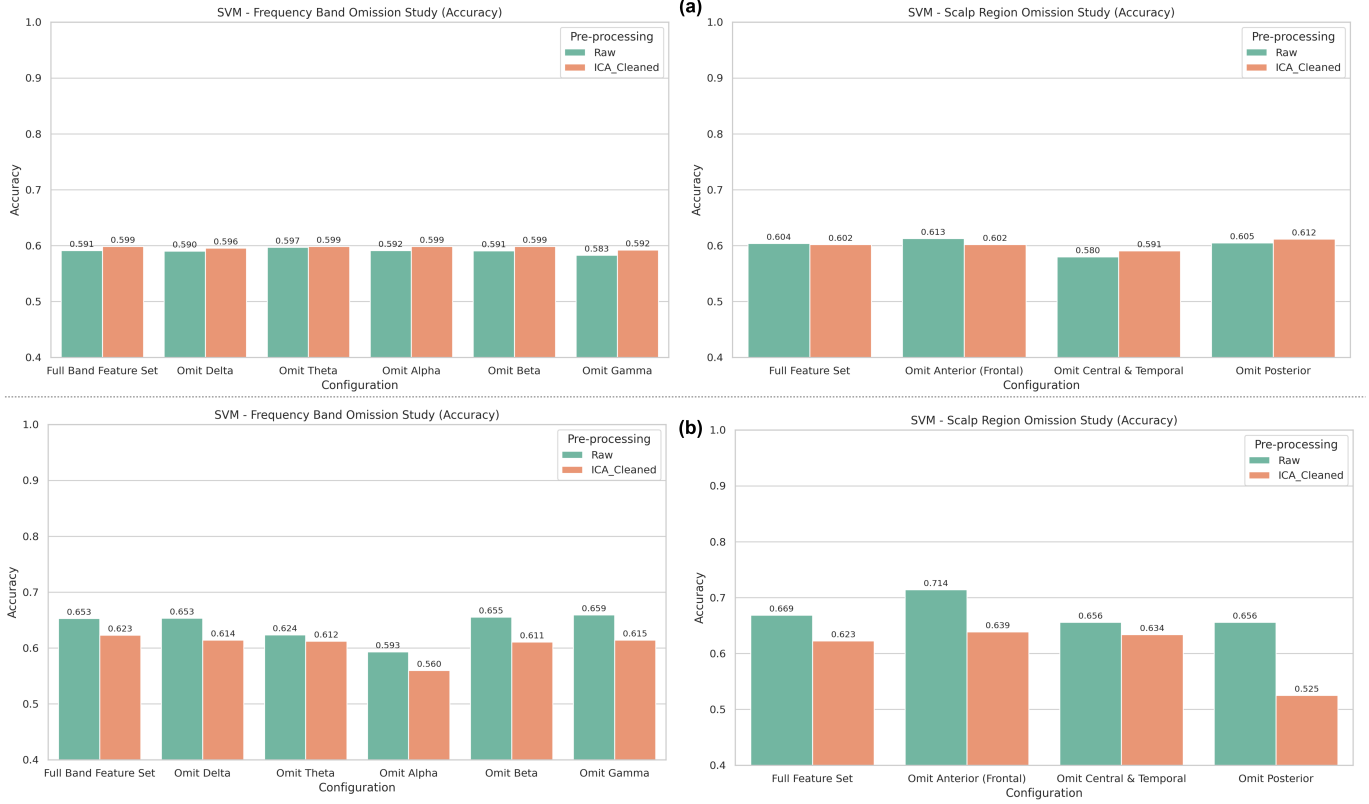


Fig. 3. to be decided

TABLE III
STATISTICAL COMPARISON OF FREQUENCY-BAND POWER BETWEEN MIND-WANDERING AND FOCUSED CONDITIONS. p -VALUES CORRESPOND TO THE MW MINUS FOCUS COMPARISON.

Band	Dataset 1			Dataset 2		
	Estimate	p -value	Decision	Estimate	p -value	Decision
Delta	0.0948	0.0901	NS	0.0992	0.1271	NS
Theta	0.0816	0.1775	NS	0.0488	2.90×10^{-7}	Significant
Alpha	-0.1113	0.1983	NS	0.1468	0.0110	Significant
Beta	-0.0494	0.4445	NS	-0.0094	0.5852	NS
Gamma	-0.0322	0.6639	NS	-0.0305	0.0733	NS

TABLE IV
RELATIVE FEATURE IMPORTANCE CONTRIBUTED BY INDIVIDUAL FREQUENCY BANDS.

Frequency Band	Dataset 1 (%)	Dataset 2 (%)
Delta	11.54	8.57
Theta	32.47	26.02
Alpha	27.94	40.60
Beta	11.24	13.19
Gamma	12.07	11.62
Other	4.73	0.00
Total	100.00	100.00

feature family in both datasets, while Hjorth parameters showed high information density and phase-based functional connectivity contributed comparatively little despite its large feature dimensionality. The ablation analysis additionally re-

TABLE V
FEATURE-FAMILY CONTRIBUTION BASED ON AGGREGATED FEATURE IMPORTANCE FOR DATASET 1 AND DATASET 2.

Feature Family	Dataset 1		Dataset 2	
	No. Features	Importance (%)	No. Features	Importance (%)
Band-power features	190	50.42	640	41.50
Hjorth features	38	27.12	128	7.03
Time-domain statistics	133	0.30	448	22.99
Oscillatory-envelope dynamics	152	3.28	512	21.26
Entropy and complexity	57	8.15	192	4.03
Functional connectivity / synchrony	361	2.39	4096	3.19
Spectral summary features	38	8.35	128	0.00

vealed dataset-dependent spatial and spectral contributions, with Theta playing a prominent role in Dataset 1 and Alpha emerging as a key driver in Dataset 2, while Central–Temporal and Posterior regions provided strong spatial discriminability. Overall, the findings demonstrate that mind wandering is characterized by reproducible low-frequency spectral dynamics and nonlinear spatial–spectral interactions, while also revealing important differences associated with EEG montage and experimental context. These results support the use of adaptive multiscale EEG analysis and nonlinear machine learning for more robust characterization of transitions between focused attention and internally directed cognitive states.

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